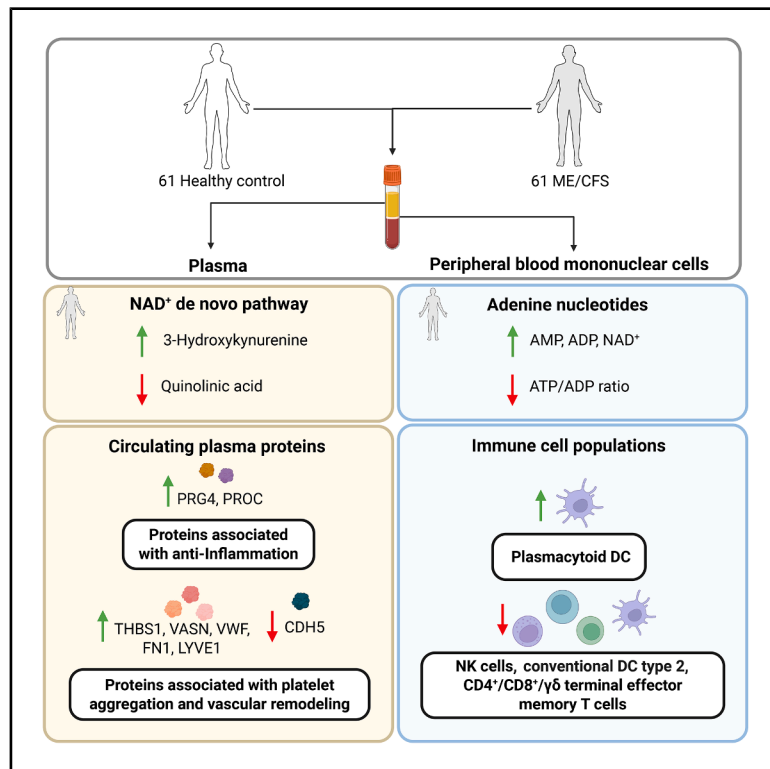


Mapping the complexity of ME/CFS: Evidence for abnormal energy metabolism, altered immune profile, and vascular dysfunction

Graphical abstract



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In brief

Heng et al. apply a multi-omics approach combining analyses of energy metabolism, immune cell composition, and plasma proteomics. They reveal altered energy metabolism and immune profiles, alongside plasma proteins associated with vascular dysfunction in ME/CFS, highlighting coordinated dysregulation across multiple biological systems in this disorder.

Highlights

- This study involves age- and sex-matched 61 healthy controls and 61 ME/CFS patients
- Abundance of AMP and ADP, with reduced ATP/ADP ratio in immune cells
- Lower proportion of terminal effector memory T cell, NK, and DC subpopulations
- Elevated plasma proteins associated with vascular and anti-inflammation activity



Article

Mapping the complexity of ME/CFS: Evidence for abnormal energy metabolism, altered immune profile, and vascular dysfunction

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SUMMARY

Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) is a complex disorder with undefined mechanisms, no diagnostic tools and treatments. To investigate concurrent system dysfunctions, we recruited age- and sex-matched ME/CFS patients and healthy controls for a multimodal analysis of energy metabolism, immune profiles, and plasma proteomics. Immune cells from ME/CFS patients show elevated adenosine monophosphate (AMP) and adenosine diphosphate (ADP) with a reduced ATP/ADP ratio, indicating decreased ATP generation and cellular energy stress. Immune profiling reveals skewing toward less mature effector subsets of CD4⁺, CD8⁺, and $\gamma\delta$ T cells, with reduced CD1c⁺CD141⁺ conventional DC type 2 and CD56^{low}CD16⁺ terminal natural killer cells. Elevated levels of plasma proteins associated with thrombus formation and vascular reactivity may contribute to the endothelial dysfunction observed in ME/CFS patients. Classification and regression tree modeling identifies variables with strong predictive potential for ME/CFS. Together, this study provides insights into the somatic symptoms and underlying biology of ME/CFS.

INTRODUCTION

Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) affects approximately 1% of the global population. According to the Canadian Consensus Criteria,¹ the key clinical presentations for ME/CFS include fatigue, post-exertional malaise, sleep dysfunction, and pain.^{2,3} Despite identification of these key symptoms, the diagnosis of ME/CFS remains challenging. Current diagnosis relies on self-reporting and thorough medical history evaluation, highlighting the need to better understand the biology of ME/CFS and identify reliable parameter(s) for patient management.

While the intrinsic heterogeneity of ME/CFS may contribute to contradictory observations,⁴ a key mechanism strongly associated with the prominent symptom of fatigue is disturbance of mitochondrial respiration.⁴ Disruption to mitochondrial metabolism leads to redox imbalance and reduced adenosine triphosphate (ATP) production.^{5–7} The kynurenine pathway

(KP), a crucial nicotinamide adenine dinucleotide (NAD⁺) *de novo* synthesis pathway, is also emerging as an important factor in ME/CFS pathology.^{8,9} NAD⁺ is critical in mitochondrial respiration as an acceptor of hydride generated in various metabolic processes such as the tri-carboxylic acid (TCA) cycle.

Mitochondrial dysfunction affects lymphocytes and myeloid cells,^{6,10,11} leading to exhaustion or hyperactivation of the immune system.¹² ME/CFS patients show altered immune profiles, including significantly lower proportions of effector T cell subsets and NK cell populations, alongside elevated levels of proinflammatory cytokines.¹¹ Nonetheless, there are also contradictory findings regarding immune cell populations and responses.¹³ An altered immune profile in ME/CFS patients is consistent with limited immune responses and sustained ineffective specific immunity. This observation shifts the focus onto professional antigen-presenting cells such as dendritic cells (DCs) and macrophages, which have remained largely uncharacterized in ME/CFS to date. An understanding of individual immune



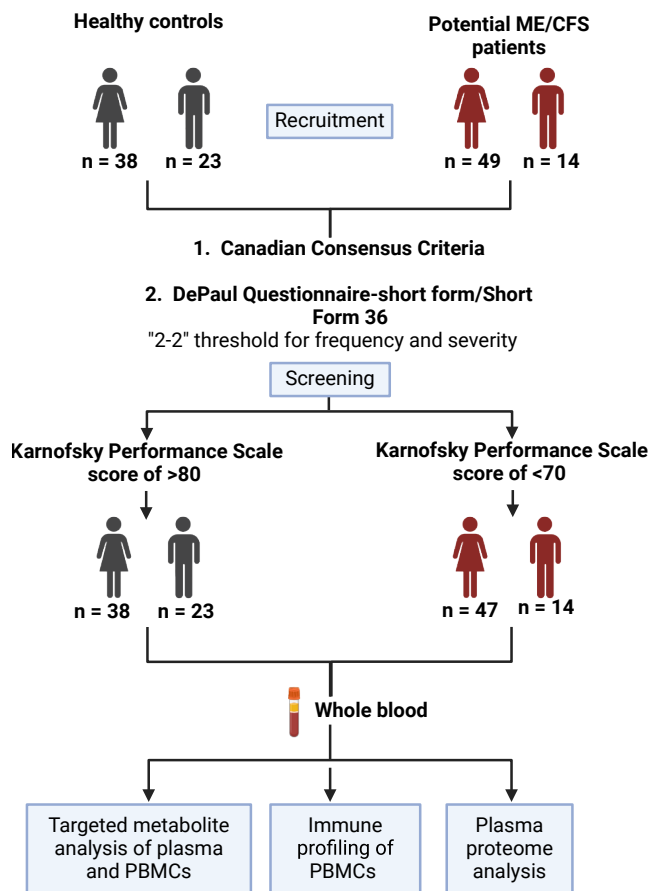


Figure 1. Overall recruitment and experimental workflow

Sixty-three potential ME/CFS patients were recruited and underwent detailed case reviews. Among them, two did not meet the Canadian Consensus Criteria¹ or satisfy the case definition classification according to the DePaul Symptoms Questionnaire-Short Form (DSQ-SF)¹⁴ and were therefore excluded. Sixty-one healthy controls were recruited separately to age- and sex-match the patient cohort. All participants were further assessed for their health status and quality of life using the SF-36 and Karnofsky Performance Scale.¹⁵ Blood was collected and processed for plasma, and peripheral blood mononuclear cells (PBMCs) were isolated for downstream targeted metabolite analysis, immune profiling, and plasma proteome analysis.

cell phenotype distributions may shed light on their functional state.

Although mitochondrial dysfunction and altered immune profile in ME/CFS patients have been extensively studied, their concurrence and biological significance remain unclear, as previous studies have often focused on a single analytical platform. This study addresses that gap by evaluating energy metabolism (NAD⁺ and ATP) together with immune cell composition in plasma and circulating immune cells from a well-defined, age- and sex-matched cohort of 61 ME/CFS patients and 61 healthy participants. Comprehensive plasma proteomics were performed to identify potential additional disease mechanisms. These analyses were conducted using a multimodal approach that combined chromatography- and mass-spectrometry (MS)-based methods together with both conventional and spectral

flow cytometry, enabling parallel assessment of metabolic, immune, and proteomic domains. Our results revealed abnormal NAD⁺ metabolism and reduced ATP generation, an altered immune profile, and a plasma protein composition associated with vascular dysfunction in ME/CFS patients. Together with exploratory machine learning on combined data from various modalities, these findings offer important insights into potential links between biochemical, immune, and vascular dysfunction in ME/CFS. Moreover, we identified variables with strong predictive potential for ME/CFS, pending further validation in an independent cohort of ME/CFS patients.

RESULTS

Study cohort demographics

Sixty-three potential ME/CFS patients were recruited and underwent detailed case review. Of these patients, 61 met the Canadian Consensus Criteria¹ and satisfied the case definition classification according to the DePaul Symptom Questionnaire-Short Form (DSQ-SF)¹⁴ for ME/CFS. Sixty-one healthy controls were recruited separately to age- and sex-match the patient cohort. All participants were further assessed on their health status and quality of life using the SF-36 and Karnofsky Performance Scale.¹⁵ The workflow is outlined in Figure 1, and the demographic and clinical features of all study participants are summarized in Table 1.

Based on the Karnofsky Performance Scale, 19.7% of ME/CFS patients were rated as disabled (score <50), another 36.1% required considerable or occasional assistance (score 50–60), and only 44.3% of patients were capable of self-care with limited normal activity (score 70) (Table 1). All patients with ME/CFS had experienced CFS for at least 5 years, with almost 46% living with the condition for 10–19 years, another 34% for 20–29 years, and 8.2% for over 30 years (Table 1). The scores for all of the SF-36 items were significantly lower in the ME/CFS patient cohort than in the healthy controls. The mean scores for role limitations due to physical health and energy/fatigue were particularly low with a value of less than 20, congruent with values observed in another study of severe ME/CFS patients.¹⁶ The physical function of the patient cohort in the present study was below 50% (score 37.2 ± 20.5), which is comparable to other ME/CFS studies, such as the metabolic study by Hoel et al. (29.3 ± 19.5)¹⁷ and the RituxME clinical trial (NCT02229942; score 35.2).¹⁸

Individuals in the control cohort were unlimited in their capacity for self-care, unaided in work activities, and either completely healthy or minimally affected by other conditions (Table 1). Importantly, the sex distribution, the mean age, and the age range were not statistically significantly different from ME/CFS patients and thus a robust foundation for comparison.

Plasma analysis reveals disruption of the NAD⁺ *de novo* pathway with accumulation of 3-hydroxykynurenine

Given that the KP is the major NAD⁺ *de novo* synthesis pathway in mammals (Figure 2A), we measured KP metabolites in plasma from healthy controls and patients with ME/CFS. We observed a deviation at the first node of this pathway at which kynurenine (KYN) can be converted to 3-hydroxykynurenine (3HK),

Table 1. Demographics and quality of life of healthy controls and ME/CFS patients

	Healthy controls (n = 61)	Patients (n = 61)	p value
Age in years (mean; range)	45.87 (26–76)	49.11 (20–79)	0.3129
Sex (F/M; % female)	38/23; 62.0%	47/14; 77.0%	0.1151
Duration of illness (years; mean $\pm$ SD)	NA	17.6 $\pm$ 8.3	–
Less than 10 years	NA	11.5%	–
10–19 years	NA	45.9%	–
20–years	NA	34.4%	–
30–40 years	NA	8.2%	–
Karnofsky Performance Scale index (%)			
30: severely disabled; hospital admission is indicated although death not imminent	0%	3.3%	–
40: disabled; requires special care and assistance	0%	16.4%	–
50: requires considerable assistance and frequent medical care	0%	16.4%	–
60: requires occasional assistance but is able to care for most of his/her personal needs	0%	19.7%	–
70: cares for self; unable to carry on normal activity or to do active work	0%	44.3%	–
80: normal activity with effort; some signs or symptoms of disease	4.9%	0%	–
90: able to carry on normal activity; minor signs or symptoms of disease	29.5%	0%	–
100: normal; no complaints; no evidence of disease	65.6%	0%	–
Quality of life (SF-36 score; mean $\pm$ SD)			
Physical functioning	93.4 $\pm$ 9.4	37.2 $\pm$ 20.5	<0.0001
Role limitations due to physical health	80.6 $\pm$ 35.2	4.9 $\pm$ 15.7	<0.0001
Role limitations due to emotional problems	86.3 $\pm$ 28.8	37.1 $\pm$ 42.7	<0.0001
Energy/fatigue	72.6 $\pm$ 27.3	13.7 $\pm$ 15.8	<0.0001
Mental health/emotional well-being	84.1 $\pm$ 17.8	52.5 $\pm$ 21.4	<0.0001
Social functioning	88.4 $\pm$ 18.1	26.0 $\pm$ 21.1	<0.0001
Pain	88.0 $\pm$ 15.4	36.1 $\pm$ 24	<0.0001
General health	76.3 $\pm$ 28.0	21.2 $\pm$ 14.2	<0.0001
Health change	73.4 $\pm$ 27.7	43.4 $\pm$ 25	<0.0001

kynurenic acid (KYNA), or anthranilic acid (AA). A higher concentration of 3HK in ME/CFS patients (34.29 nM; 95% confidence interval [CI] 31.26–37.22) was measured compared to healthy controls (26.67 nM; 95% CI 25.09–32.36) (Figure 2B), and the levels of KYNA were lower (23.38 nM; 95% CI 22.4–33.05) than in healthy control samples (30.26 nM; 95% CI 29.92–42.41; Figure 2B), consistent with elevated kynurenine monooxygenase (KMO) activity.¹⁹ Furthermore, we observed lower levels of picolinic acid (PIC) and quinolinic acid (QUIN) in ME/CFS patients (PIC: 141.8 nM; 95% CI 71.64–1297 and QUIN: 412.5 nM; 95% CI 402.6–532) as compared to healthy controls (PIC: 208 nM; 95% CI 193–290.4 and QUIN: 463.5 nM; 95% CI 463.7–571) (Figure 2B). No differences were evident in the median levels of tryptophan (TRP), KYN, AA, and 3-hydroxyanthranilic acid (3HAA) between the two cohorts (Figure 2B). The net effect of these changes reduces systemic availability of QUIN, the substrate for NAD⁺ generation.

High concentration of AMP in plasma of ME/CFS patients

We next investigated whether the KP dysregulation in ME/CFS patients could contribute to energy deficiency. It is well established that NAD⁺ plays a critical role in ATP production.²⁰ There-

fore, we analyzed plasma concentrations of adenine nucleotides, such as adenosine monophosphate (AMP), adenosine diphosphate (ADP), and ATP in the plasma of ME/CFS patients and healthy controls. Although we observed higher median levels of AMP in ME/CFS patients (312.2 nM; 95% CI 285.0–439.7) than in healthy controls (147.2 nM; 95% CI 129.8–432.7), this difference did not achieve statistical significance (Figure 2C). There was no difference in the levels of ADP or ATP between the cohorts. Only the median levels of AMP were higher in female patients compared to female healthy controls (Figure S1).

High metabolic and redox potential profile in PBMC of ME/CFS patients

Alterations in the immune system are frequently observed in ME/CFS, and since immune responses are closely associated with cellular energy states²¹ and NAD⁺ is synthesized and compartmentalized intracellularly, we evaluated the energy metabolism of patient peripheral blood mononuclear cells (PBMCs). Elevated levels of NAD⁺ (300.3 nM; 95% CI 248.3–311.9 vs. 257.7 nM, 95% CI 219.3–274) and higher NADP⁺/NADPH ratio (0.023, 95% CI 0.02–0.06 vs. 0.016, 95% CI 0.014–0.023) were measured in patients compared to healthy controls (Figure 2D).

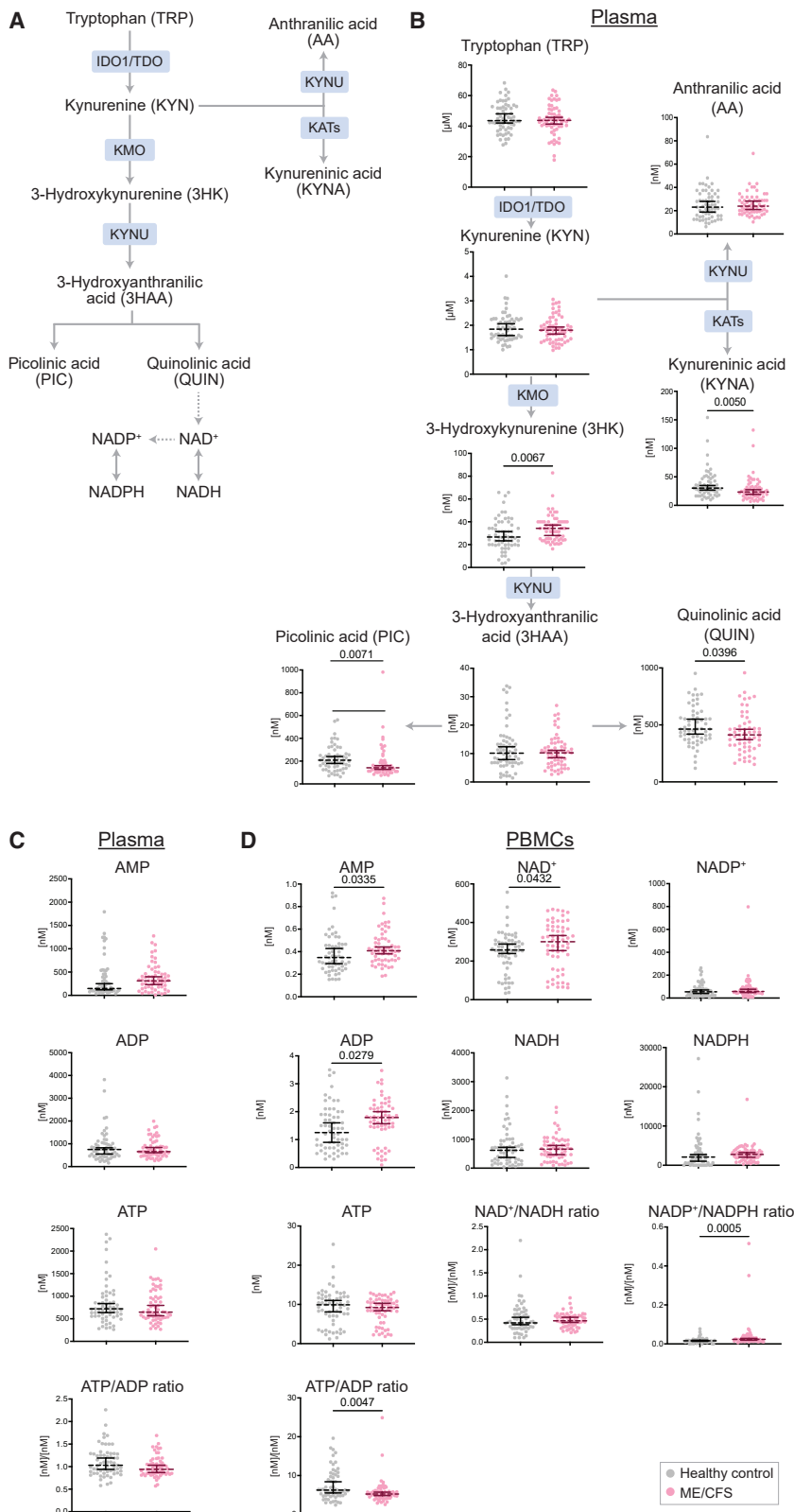


Figure 2. Analysis of energy metabolism of peripheral and immune cell population

(A) Simplified schematic of the kynurenine pathway (KP), illustrating its role in producing NAD⁺.

(B) Shows plasma levels of metabolites in the KP. 3HK was significantly higher, whereas KYNA, PIC, and QUIN were significantly lower in patients compared to healthy controls.

(C) ME/CFS patients had a higher level of AMP and lower ATP/ADP ratio in their plasma, indicating reduced ATP production.

(D) In ME/CFS patient PBMCs, levels of NAD⁺, AMP, ADP, and the NADP/NADPH ratio were elevated, whereas the ATP/ADP ratio was reduced. Collectively, these findings suggest abnormal energy production in patients with ME/CFS, potentially contributing to their chronic fatigue. Each scatterplot represents individual participants (dots), and data are presented as the median and 95% confidence intervals. *p* value was calculated using Mann-Whitney U test. Plasma samples were collected from 61 ME/CFS patients and 61 healthy controls. However, matched PBMC samples were available for only 60 patients and 60 healthy controls due to insufficient immune cell yield from one healthy control sample and no quantification signals from one ME/CFS patient sample, as described under sample collection. For the calculation of the NADP/NADPH ratio, only 47 of 60 healthy controls were included, as 13 controls recorded zero value for NADP⁺.

Moreover, in contrast to the findings in plasma, the levels of AMP (0.4 nM, 95% CI 0.39–0.47) and ADP (1.79 μ M, 95% CI 1.49–1.89) were significantly higher in ME/CFS patients than in controls (AMP: 0.35 nM, 95% CI 0.34–0.45; ADP: 1.25 μ M, 95% CI 1.19–1.6). The dynamics of the adenylate energy system were determined by calculating the rate of ATP synthesis from phosphorylation of ADP, using the ATP/ADP ratio.²² We observed a reduced ATP/ADP ratio in ME/CFS patients (5.25, 95% CI 4.93–5.71 cf. 6.25, 95% CI 5.63–7.75) (Figure 2D). Although there were no significant differences in plasma adenosine levels, Spearman's correlation analysis revealed a negative association between PBMC ATP and plasma AMP ($p = 0.0497$). A trend-level positive association was also observed between PBMC AMP and plasma ATP ($p = 0.073$) although this did not reach statistical significance (Table S1). Among the female patients, there were significantly lower levels of ATP and a reduced ATP/ADP ratio compared to healthy controls. There was a trend toward increased levels of AMP in female ME/CFS patients (Figure S1).

Taken together, the metabolomic analysis revealed two key features: 3HK accumulation, which indicates disruption of the KP and is linked to NAD⁺ production, and higher levels of AMP implying compromised energy status at a systemic level. These findings align with the survey data showing 82% of patients report chronic fatigue (Table 1)—a classic symptom of ME/CFS.⁴ Additionally, the increased levels of AMP and ADP, along with a reduced ATP/ADP ratio in PBMC suggest lower energy generation in immune cell populations.

Altered CD11c⁺HLA-DR⁺ dendritic cell populations in ME/CFS

To understand how perturbations in energy metabolism may be associated with changes in immune cell populations in ME/CFS, PBMCs were immunophenotyped using high-dimensional spectral flow cytometry. Although the total CD11c⁺HLA-DR⁺ DC population showed no significant difference between ME/CFS patients and healthy control PMBCs, further analysis revealed that there was a significantly lower proportion of CD1c⁺CD141[−] conventional DC type 2 (cDC2; $p = 0.001$) but not CD1c[−]CD141⁺ conventional DC type 1 (cDC1) and higher proportion of CD11c[−]CD123⁺ plasmacytoid DC (pDC; $p = 0.0149$) in patient PBMCs (Figure 3A). Within the female cohorts, ME/CFS patients had a significantly lower proportion of cDC2 ($p = 0.001$). There were no detectable differences in CD14⁺ monocyte populations between female patients and the healthy control cohort (Figure S2).

Reduced mature effector subsets among the peripheral blood lymphocytes of ME/CFS patients

Peripheral blood T cells were divided into subsets reflecting their maturation level and function based on surface antigen expression. Within the CD4⁺ T cell population, there were significantly less CD45RA[−]CCR7[−] effector memory CD4⁺ T cells in ME/CFS patients compared to healthy controls ($p = 0.0491$; Figure 3B), but no difference was observed in the proportions of other CD4⁺ T cell subsets (Figure S2). Recognizing the pivotal role of the T cell response in protecting against infections, we further analyzed the status of each CD4⁺ T cell effector memory subset. The CD45RA[−]CCR7[−]CD4⁺ effector memory T cell pop-

ulation was dominated by cells expressing the CD27⁺CD28⁺ early effector memory phenotype ($p = 0.0037$), while the CD27[−]CD28[−] terminal effector memory subset was significantly reduced in patient PBMCs when compared to those from healthy controls ($p = 0.0056$) (Figure 3B).

The general proportions of CD8⁺ T cell subsets were found to mirror those of CD4⁺ cells in ME/CFS patients. A significantly lower proportion of CD45RA[−]CCR7[−] effector memory CD8⁺ T cells ($p = 0.019$) and particularly the CD27[−]CD28[−] terminal effector memory subset ($p = 0.0476$) was evident in ME/CFS patients. Instead, there was a higher proportion of cells bearing the CD27⁺CD28⁺ early effector memory phenotype (Figure 4C). Similar differences in phenotype were noted in the $\gamma\delta$ T cell population. The proportion of CD45RA⁺CCR7[−] TEMRA $\gamma\delta$ T cells was significantly lower in ME/CFS patients ($p = 0.0397$) (Figure 3D).

We observed a smaller fraction of total CD56⁺ NK cells in patients compared to healthy controls, and this was primarily attributable to the reduction of CD56^{low}CD16⁺ terminal NK cells in these patients ($p = 0.0187$; Figure 3E). There was no difference in subsets of B cells or regulatory T cells between ME/CFS patients and healthy controls (Figure S2). Female patients had a significantly reduced proportion of CD27[−]CD28[−]CD4⁺ terminal effector memory ($p = 0.0158$) and CD27[−]CD28[−]CD8⁺ intermediate effector memory ($p = 0.0451$) T cell subsets and CD56^{low}CD16⁺ terminal NK cells ($p = 0.0324$) compared to healthy controls (Figure S2).

Multiple plasma cytokines (interleukins [IL]-1 β , IL-2, IL-4, IL-6, IL-8, IL-10, IL-12p70, IL-13, IL-17A, IL-18, IL-23, IL-33, monocyte chemoattractant protein 1[MCP-1], tumor necrosis factor [TNF], interferon alpha [IFN- α], and IFN- γ) were detected and found to be similar in abundance in both cohorts (Figure S3).

Altered levels of endothelial and circulating proteins in the plasma of patients with ME/CFS

We expanded our analysis to include a more comprehensive range of plasma proteins using MS-based proteomics. This allowed us to identify and quantify a total of 322 distinct proteins with high confidence ($\leq 1\%$ false discovery rate, ≥ 1 unique peptides) throughout all patient and control plasma samples. Among the 322 quantified plasma proteins, 20 proteins demonstrated significantly upregulated expression in the ME/CFS patient cohort compared to healthy controls, while eight proteins were significantly downregulated (p value < 0.05 and a fold change ± 1.2 ; Figure 4A; Table 2). Most of the upregulated proteins were secreted proteins, including fibronectin 1 (FN1), sex-hormone-binding globulin (SHBG), protein C (PROC), proteoglycan 4 (PRG4), thrombospondin 1 (THBS1), von Willebrand factor (VWF), epidermal growth factor (EGF) containing fibulin extracellular matrix protein 1 (EFEMP1), and fetuin B (FETUB). Circulating immune-related proteins included beta-2-microglobulin (B2M), complement-factor-H-related 5 (CFHR5), complement-factor-H-related 2 (CFHR2), mannose-binding lectin (MBL)-associated serine protease 1 (MASP1), polymeric immunoglobulin receptor (PIGR), immunoglobulin heavy variable 5-51 (IGHV5-51), immunoglobulin heavy variable 3-74 (IGHV3-74), and immunoglobulin kappa variable 6-21 (IGKV6-21).

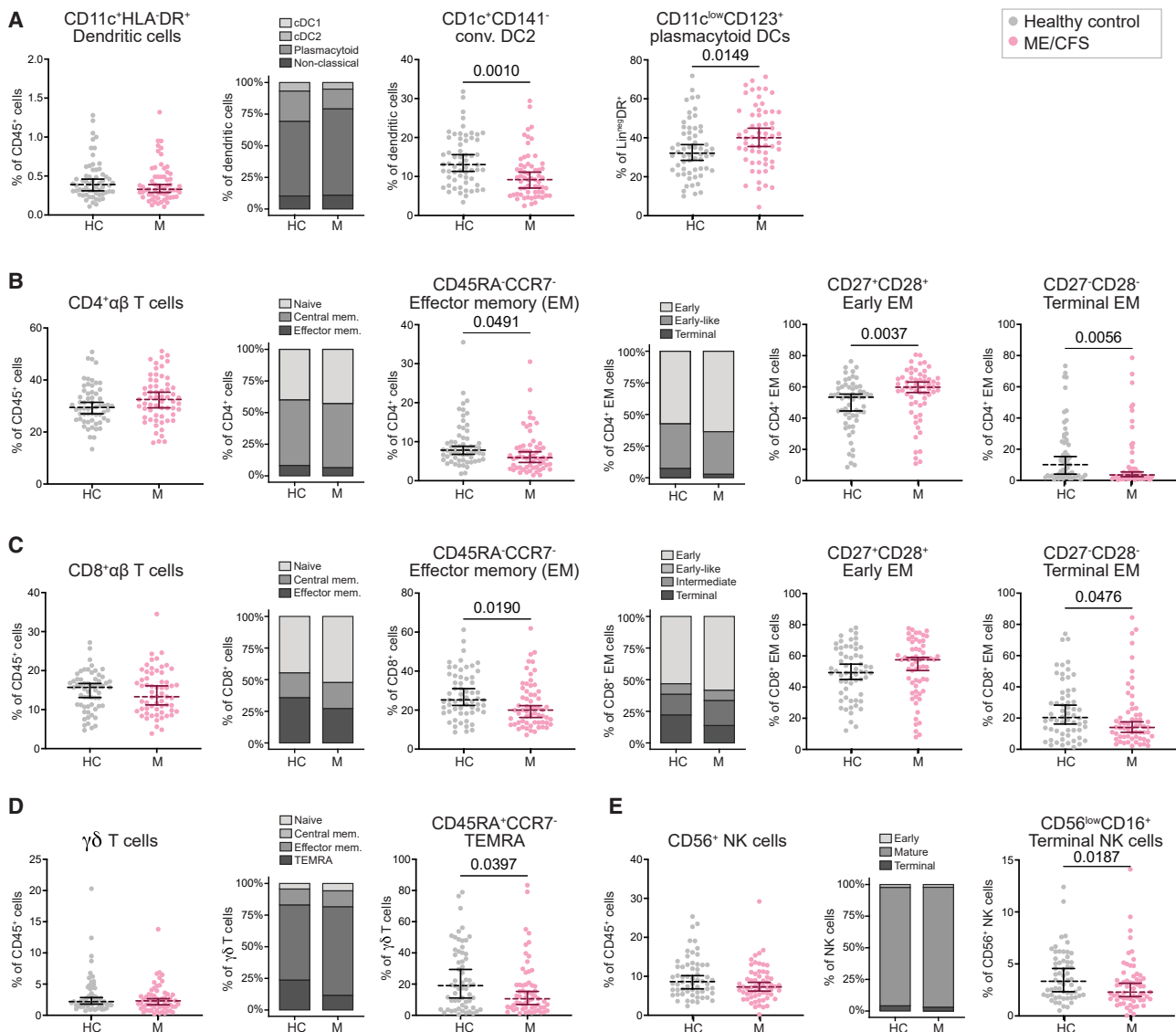


Figure 3. Immunophenotyping of peripheral blood mononuclear cells

We observed a significantly lower proportion of (A) $CD11c^+CD141^-$ conventional dendritic cells 2 (cDC2) and a higher proportion of $CD11c^+CD123^+$ plasmacytoid dendritic cells (pDC) in ME/CFS patients. There was a significantly lower proportion of effector memory cell subsets of (B) $CD4^+$ and (C) $CD8^+$ T cells, primarily attributable to the reduction in $CD27^-CD28^-$ terminal subset of the effector memory population. Additionally, ME/CFS patients also had decreased proportions of (D) the $CD45RA^+CCR7^-$ TEMRA $\gamma\delta$ T cell subset and (E) $CD56^{low}CD16^+$ terminal NK cells. Collectively, the reduction in pathogen-responsing immune cell subsets suggests an altered immune system in ME/CFS patients and an increased vulnerability to viral infection. Each scatterplot represents individual participants (dots), and data are presented as the median and 95% confidence intervals. *p* value was calculated using Mann-Whitney U test or Welch's *t* test. PBMCs were collected from 61 ME/CFS patients and 60 age- and sex-matched healthy controls.

Lymphatic vessel endothelial hyaluronan receptor 1 (LYVE1)²³ and vasorin (VASN)²⁴ are membrane-bound proteins that can be enzymatically cleaved to release their ectodomains into circulation. In contrast, DnaJ heat shock protein family 40 member C2 (DNAJC2) and prenylcysteine oxidase 1 (PCYOX1) are intracellular proteins with no evidence supporting their secretion or release. Among the upregulated proteins (Figure 4B), levels of LYVE1 and DNAJC2 were significantly elevated with greater than or equal to 2-fold increase in ME/CFS patient plasma, whereas THBS1, VASN, IGHV3-74, IGKV6-21, PIGR, CFHR5,

and EFEMP1 were increased by 1.6- to 1.9-fold in the plasma of ME/CFS patients. Levels of the remaining proteins SHBG, PRG4, PROC, FN1, B2M, PCYOX1, VWF, CFHR2, MASP1, IGHV5-51, and FETUB were also elevated in ME/CFS patients' plasma, with fold change between 1.2 and 1.5 compared to healthy controls (Figure 4B).

Six of the eight significantly downregulated proteins—immunoglobulin lambda like polypeptide 1 (IGLL1), immunoglobulin heavy constant gamma 2 (IGHG2), immunoglobulin heavy constant gamma 3 (IGHG3), immunoglobulin kappa variable

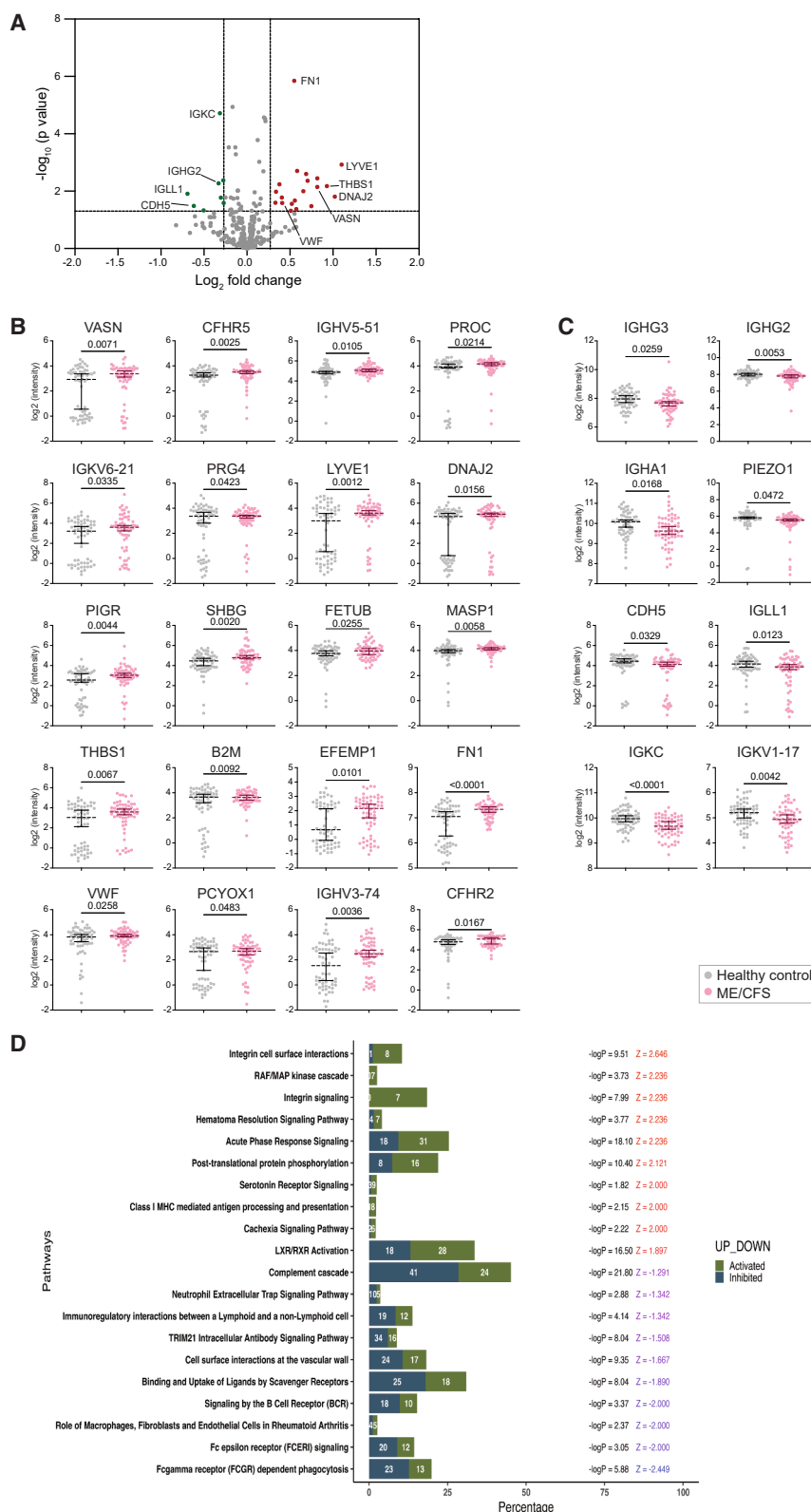


Figure 4. Proteomic analysis of plasma protein in ME/CFS patients and healthy controls

(A) Volcano plot illustrating the distribution of plasma proteins that were shown to be differentially expressed in ME/CFS patients compared to healthy controls. Twenty proteins were elevated in ME/CFS patients: namely (B) LYVE1, DNAJ2, THBS1, VASN, IGHV3-74, IGV6-21, PIGR, CFHR5, EFEMP1, SHBG, PRG4, PROC, FN1, B2M, PCYOX1, VWF, CFHR2, MASP1, IGHV5-51, and FETUB.

(C) Eight plasma proteins were significantly lower in ME/CFS patients: IGHG3, IGHG2, IGKV1-17, IGHA1, IGKC, PIEZO1, CDH5, and IGLL1. Many of these proteins are associated with immune activity or vascular integrity. Each scatterplot represents individual participants (dots), and data are presented as the median and 95% confidence intervals. A relative abundance comparison was made by *t* test in R. Proteins with *p* value <0.05 and fold change >1.2 were determined to be differentially expressed between healthy controls and patients with ME/CFS. Plasma was collected from age- and sex-matched 61 ME/CFS patients and 61 healthy controls.

(D) Top 10 upregulated and downregulated pathways in MECFS patient compared to controls, ranked by the highest and lowest Z score based on differentially expressed proteins. The bars are color-coded to indicate the regulatory status of the genes within each pathway.

Table 2. Differentially expressed plasma proteins between ME/CFS and healthy controls

Gene name	Protein name	Cell origin (most abundantly produced)	Mechanism of release	Known function as soluble form	Fold-change
LYVE1	lymphatic vessel endothelial hyaluronan receptor 1	lymphatic endothelial cells	enzymatically cleaved, soluble circulating shed protein	binds to fibroblast growth factor 2, preventing lymphangiogenesis ²³	2.1
DNAJC2	DnaJ heat shock protein family 40 member C2	all cell types	intracellular (cytoplasm), no known evidence of extracellular release	not applicable	2
THBS1	thrombospondin	endothelial cells, platelets	soluble circulating shed protein	- soluble THBS1 binds to CD36 and CD47 on endothelial cells, inhibiting NO signaling pathway and angiogenesis²⁵ - binds to CD36 and CD47 on DCs to reduced cytokine release and phenotypic maturation²⁶ - promotes platelet aggregation at site of vascular injury in the presence of VWF, leading to formation of thrombus^{27,28}	1.9
IGHV3-74	immunoglobulin heavy variable 3-74	immune cells	soluble circulating secreted protein, as part of antibodies	antigen recognition and binding region of a soluble antibody	1.8
VASN	vasorin	vascular smooth muscle cells of blood vessels	enzymatically cleaved, soluble circulating shed protein	soluble VASN binds to TGF- β receptor, inhibiting TGF- β pathways and angiogenesis ²⁴	1.8
IGKV6-21	immunoglobulin kappa variable 6-21	immune cells	soluble circulating secreted protein, as part of antibodies	facilitate antigen binding	1.7
CFHR5	complement factor H-related 5	immune cells	soluble circulating secreted protein	a negative regulator of the major complement pathway proteins ^{29,30}	1.6
EFEMP1	EGF containing fibulin extracellular matrix protein 1	fibroblast	soluble circulating protein	organization of extracellular matrix ³¹	1.6
PIGR	polymeric immunoglobulin receptor	immune cells	membrane-bound receptor and soluble shed protein	stabilizes antibodies in secretions ³²	1.6
FN1	fibronectin 1	hepatocytes	soluble circulating protein	promotes the aggregation of platelets at site of vascular injury, leading to thrombus formation ³³	1.5
PRG4	proteoglycan 4	synoviocytes	soluble circulating protein	anti-inflammatory ^{34,35}	1.5
PROC	protein C	hepatocytes	soluble circulating protein	anti-inflammatory ³⁶	1.5
SHBG	sex-hormone-binding globulin	hepatocytes	soluble circulating protein	regulates bioavailability of sex hormones ³⁷	1.5
B2M	beta-2-microglobulin	all nucleated cells	soluble circulating shed protein	released as part of its routine degradation and recycling of MHC class I molecules	1.4
PCYOX1	prenylcysteine oxidase 1	all cell types	intracellular (cytoplasm), no known evidence of extracellular release	not applicable	1.4

(Continued on next page)

Table 2. Continued

Gene name	Protein name	Cell origin (most abundantly produced)	Mechanism of release	Known function as soluble form	Fold-change
CFHR2	complement factor H-related 2	immune cells	soluble circulating secreted protein	a negative regulator of the major complement pathway proteins ^{29,30}	1.3
FETUB	fetuin B	hepatocytes	soluble circulating protein	supports fertilization of oocytes	1.3
IGHV5-51	immunoglobulin heavy variable 5-51	immune cells	soluble circulating secreted protein, as part of antibodies	antigen recognition and binding region of a soluble antibody	1.3
MASP1	MBL-associated serine protease 1	immune cells	soluble circulating secreted protein	binds to pattern recognition molecules and activator of lectin pathway of the complement system ^{38,39}	1.3
VWF	von Willebrand factor	endothelial cells, platelets (megakaryocytes)	soluble circulating protein	promotes the aggregation of platelets at site of vascular injury, leading to thrombus formation ²⁸	1.3
IGHA1	immunoglobulin heavy constant alpha 1	immune cells	soluble circulating secreted protein, as part of antibodies	a component of the constant region of the heavy chain of IgA	–1.2
IGHG3	immunoglobulin heavy constant gamma 3	immune cells	soluble circulating secreted protein, as part of antibodies	heavy constant region of a soluble antibody	–1.2
IGKC	immunoglobulin kappa constant	immune cells	soluble circulating secreted protein, as part of antibodies	heavy constant region of a soluble antibody	–1.2
IGKV1-17	immunoglobulin kappa variable 1-17	immune cells	soluble circulating secreted protein, as part of antibodies	facilitate antigen binding	–1.2
IGHG2	immunoglobulin heavy constant gamma 2	immune cells	soluble circulating secreted protein, as part of antibodies	heavy constant region of a soluble antibody	–1.3
PIEZO1	Piezo-type mechanosensitive ion channel component 1	endothelial cells	intracellular (cytoplasm), no known evidence of extracellular release	not applicable	–1.4
CDH5	cadherin 5/VE-cadherin	endothelial cells	enzymatically cleaved, soluble circulating shed protein	no known evidence of function	–1.5
IGLL1	immunoglobulin lambda like polypeptide 1	immune cells	soluble circulating secreted protein, as part of antibodies	component of the surrogate light chain in the pre-B cell receptor	–1.6

1-17 (IGKV1-17), immunoglobulin heavy constant alpha 1 (IGHA1), and immunoglobulin kappa constant (IGKC)—are circulating immune-related proteins. The remaining two downregulated proteins, Piezo-type mechanosensitive ion channel component 1 (PIEZO1) and cadherin 5/VE-cadherin (CDH5), are membrane proteins. Notably, CDH5 can be enzymatically cleaved and released to circulation under hypoxic⁴⁰ or inflammatory conditions.⁴¹ PIEZO1, CDH5, and IGLL1 showed a reduction of 1.4- to 1.6-fold, while IGHG3, IGHG2, IGKV1-17, IGHA1, and IGKC were downregulated by 1.2- to 1.4-fold (Figure 3C). These findings were recapitulated in the female patient cohort (Figure S4). Among the antibody fragments detected, IGHV3-74 and IGHV5-51 may confer specificity for particular antigens. A search in The Patent & Literature Antibody Database (PLAbDab⁴²; accessed on 20th October 2024) showed that IGHV5-51 is strongly associated with binding to human parainfluenza virus 3, a pathogen that causes severe respiratory illness.⁴³ It is noteworthy to mention that most of the top upregulated proteins are associated with vascular processes, while the top downregulated proteins are linked to immune responses. This pattern is reflected in the canonical pathway analysis (Figure 4D), where the most significantly upregulated pathways include integrin cell surface interactions, the RAF/MAP kinase cascade, hematoma resolution signaling, acute phase response signaling, and the coagulation system. Together, these pathways suggest endothelial activation and vascular remodeling, supported by the upregulation of FN1, VASN, THBS1, and VWF. In contrast, the most significantly downregulated pathways included Fc γ -receptor-mediated phagocytosis, B cell receptor signaling, and Fc ϵ RI signaling, driven by reduced levels of immunoglobulin-related proteins. These findings align with the observed lower abundance of circulating immunoglobulins. Collectively, these pathway alterations highlight a disruption of vascular and immune homeostasis in ME/CFS.

To assess the impact of filtering thresholds on protein quantification, we repeated the analysis using a stricter criterion requiring proteins to be present in at least 70% of all samples. This reduced the number of differentially expressed proteins from 28 to 23, excluding PRG4, PCYOX1, PIEZO1, IGHV3-74, and EFEMP1 (Figure S5). The 23 remaining proteins showed the same fold change as those identified using the 50% threshold, supporting the robustness of the findings. Although excluded under the 70% threshold, PRG4 and PCYOX1 remained to be in higher abundance in ME/CFS patients, while PIEZO1 showed lower abundance in ME/CFS patients compared to healthy controls. IGHV3-74 and EFEMP1 were detected in over 70% of ME/CFS patients but in fewer than 55% of controls, suggesting preferential expression in the patient group. These differences between the 50% and 70% selection threshold likely reflect how data-independent acquisition mass spectrometry (DIA-MS) handles missing values, where absence may result from detection limit rather than true lack of expression.⁴⁴ Given the heterogeneity of ME/CFS and our aim to capture both shared and condition-specific changes, we retained the 50% threshold to preserve proteins that may hold biological relevance but would otherwise be excluded under stricter filtering.

Predictive performance of altered biological parameters

Classification and regression tree (CART) modeling, a predictive algorithm used in machine learning to develop classification models, was employed to explore whether DSQ-SF-based ME/CFS diagnosis may be predicted by any combination of the 114 biological variables selected from all PBMC subsets, metabolites, and differentially expressed plasma proteins (Table S2). Sex and age were also included in the modeling process. The decision to include only the differentially expressed plasma proteins was made to reduce the number of inputs and help mitigate the risk of overfitting, thereby supporting interpretability in this initial exploratory analysis.⁴⁵ This would increase the likelihood that the model produces consistent and interpretable results during this initial exploratory analysis. In contrast, all immune cell and metabolite variables were retained, as they have been previously linked to ME/CFS. This approach was intended to balance dimensionality reduction with biological relevance and produce a simple and parsimonious model that could generate hypotheses for future validation. We identified seven parameters from among plasma proteins (FN1, VWF, LYVE1, IGHG2, and THBS1), an immune cell subset (cDC1), and plasma AMP levels to be the most significant contributors to predicting ME/CFS. Although the proportions of cDC1 were not found to be significantly different between the cohorts, its inclusion in the modeling contributes to the decision-making process through interaction with other variables. When used together as a set, the model produced the highest area under curve (96.2%), with 85.2% sensitivity, 96.7% specificity, and 91% accuracy (Figure 5). While these results are encouraging, they should be regarded as exploratory and intended to highlight potential associations.

DISCUSSION

The multimodal approach enables simultaneous investigation of multiple biological systems, potentially yielding information about interactions between them. This is of particular importance in ME/CFS, a heterogeneous disease in which different aspects of biological dysfunction are rarely studied within the same subjects. This study concurrently examines energy metabolism (NAD⁺ and ATP), immune profiles, and vascular dysfunction in a single, well-defined cohort. Our findings provide insights into the clinical and biological complexity of ME/CFS and highlight potential mechanistic interactions that may contribute to the disease's clinical manifestations.

The most striking differences in energy metabolism in ME/CFS patients include the abundance of AMP in the plasma and immune cells, alongside increased levels of ADP and reduced ATP/ADP ratio in PBMCs (Figures 1C and 1D), consistent with previous reports of reduced ATP production in PBMCs from ME/CFS patients.^{46,47} The elevated NAD⁺ levels observed in PBMCs, together with the increased AMP and ADP, suggest that NAD⁺ is not being efficiently reduced to NADH to drive ATP synthesis through oxidative phosphorylation.²⁰ This impaired reduction of NAD⁺ to NADH may also help explain the observed inverse correlation between plasma AMP and PBMC ATP levels, although this finding requires further study.

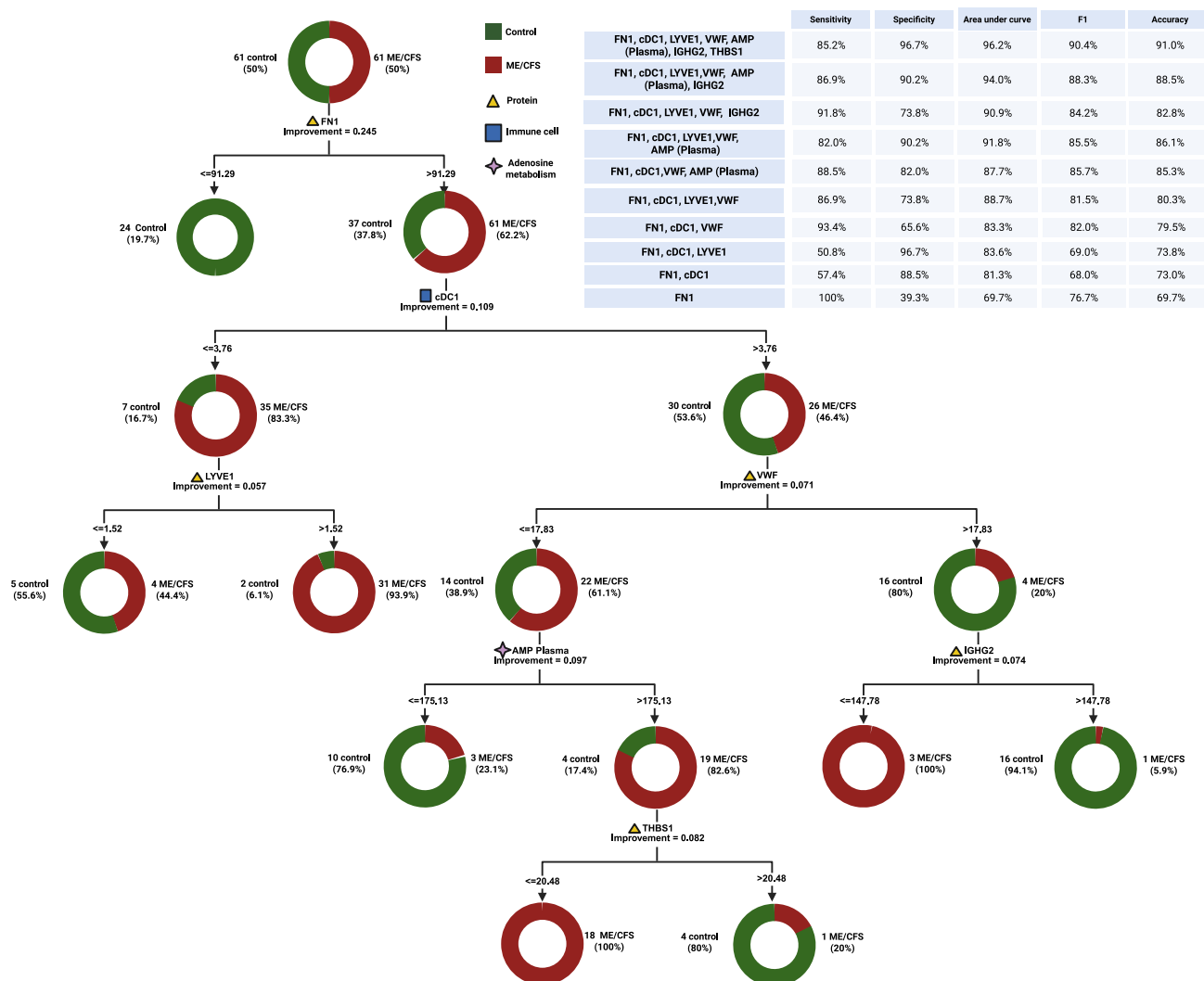


Figure 5. Identification of strong predictive potential for ME/CFS using classification and regression tree modeling

Among the 114 biological parameters, CART modeling identified seven significant contributors to discriminate between ME/CFS patients and controls with 85.2% sensitivity, 96.7% specificity, and 91% accuracy. These were FN1, cDC1, VWF, LYVE1, AMP (plasma), IGHG2, and THBS1 to predicting ME/CFS patients. CART modeling was used to develop a classification model using IBM SPSS Statistics v.29 and JMP v.17.2.

Collectively, these findings indicate mitochondrial dysfunction and energy stress.

The higher levels of plasma AMP observed in ME/CFS may be attributed to a combination of impaired purinergic metabolism involving CD39-CD73 axis,^{48,49} with insufficient conversion of AMP to adenosine by CD73, and direct extracellular release of AMP from stressed or damaged cells,⁵⁰ leading to an accumulation of AMP in the plasma. While this interpretation is consistent with known mechanisms of extracellular nucleotide regulation, further studies are required to confirm this hypothesis and establish its relevance to ME/CFS pathophysiology. Under conditions of high AMP concentration, free AMP interacts with mitochondrial fission factor and dynamin-related protein (Drp1), promoting mitochondrial fission and potentially contributing to the observed mitochondrial dysfunction.⁵¹

We also demonstrated that the KP is dysregulated with elevated levels of 3HK (implying elevated activity of KMO) and reduced levels of KYNA, PIC, and QUIN. Previous studies have described similar KP alterations in ME/CFS.^{9,17,52} 3HK induces oxidative stress and mitochondrial membrane potential loss through its generation of reactive oxygen species⁵³ (Figure 1A). While the overexpression of KMO in cells reduces sensitivity to oxidative stress,⁵⁴ it also has an effect on mitochondrial dynamics, due to its location on the outer mitochondrial membrane⁵⁵ and interaction with Drp1. This interaction leads to the dephosphorylation of Drp1 at Serine 616 and Serine 637, enhancing Drp1 GTPase activity and thereby promoting mitochondrial fission and disruption.^{56–58} This aligns with the recent findings indicating that interference with mitochondrial proteins in muscle cells can lead to increased fatigue likely due to increased endoplasmic reticulum stress mediated by

overexpression of the gene *WASF3*.⁷ Our findings suggest that the energy constraints in ME/CFS extend beyond mitochondrial dysfunction and could involve alterations in the KP. These combined changes might contribute to clinical fatigue symptoms in ME/CFS, but further studies are needed to confirm this mechanism.

Immunophenotyping showed changes in the proportions of NK and DC populations but not monocytes. Consistent with existing studies,^{59–62} we observed significantly smaller populations of CD56^{low}CD16⁺ NK cells in ME/CFS patients (Figure 3E). The lower proportion of this NK cell subtype, known for its potent cytolytic⁶³ and antiviral⁶⁴ activity, implies an increased vulnerability to viral infection.⁶⁵ Our observation of increased proportions of pDC (which specialize in responding to viral infections by producing type I and III IFNs⁶⁶) in peripheral blood of ME/CFS patients is consistent with earlier findings of pDC enrichment in ME/CFS duodenum.⁶⁷ The high frequency of pDC and the increased production of type I IFNs⁶⁸ have the potential to contribute to the increased susceptibility to autoimmune diseases in ME/CFS.⁶⁹ Moreover, we noted significantly lower proportions of cDC2 in patients compared to healthy controls. As these cells primarily respond to flagellin and bacterial proteins, this reduction may be linked to the microbiome dysbiosis observed in ME.⁷⁰ The altered cDC2 populations observed in ME/CFS suggests changes in DC composition that warrants further investigation. cDC2 are also essential cross-presenting immune cells that prime CD4⁺ and CD8⁺ T cells.⁷¹ These findings are important due to the diverse roles played by these DC subsets in enhancing innate immune cell responses and priming adaptive immune cells. cDC2 play a critical role in co-stimulation with pDC to enhance NK cell cytotoxicity and in the differentiation of plasma cells from activated B cells.⁷² This interaction potentially explains the previously observed lower NK cytotoxicity in ME/CFS. However, our findings regarding NK cells differ from those of Brenu et al., who observed a decreased population of CD56^{high}CD16⁺ NK cells and pDC but no difference in CD56^{dim}CD16⁺ NK cells or mature DCs.⁶¹ Differences were also noted when comparing T and B cell observations with other studies.^{73,74} This discrepancy may be associated with differences in the patient selection criteria; Brenu et al. used the Center for Disease Prevention and Control case definition 1994 criteria while we used the Canadian Consensus Criteria. The variation in criteria highlights the potential impact of patient selection on the observed immune cell alterations in ME/CFS.

Within adaptive immune cells, we observed a decline in the CD45RA⁺CCR7⁺ effector memory population in CD4⁺ αβ, CD8⁺ αβ, and γδ T cells, which could be attributed to various factors. First, the reduced population of cDC2, the predominant subset of conventional DC,⁶⁶ may limit T cell priming capacity.⁷⁵ A role for NK cells in antigen presentation has been described.⁷⁶ Accordingly, the decreased fraction of CD56⁺ NK cells in our patient cohort could also be reflected in reduced antigen presentation,⁷⁶ further impacting T cell activation and proliferation. The increased proportion of CD27⁺CD28⁺ early effector memory subsets of CD4⁺ and CD8⁺ T cells observed in ME/CFS patients may be associated with the lack of exposure to presented antigens, thereby limiting the ability to respond to infections.⁷⁷ One further potential explanation for the imbalance in effector

populations may stem from our proteomic finding of elevated circulating levels of LYVE1 in the plasma of ME/CFS patients. LYVE1, a major receptor for hyaluronan, is predominantly expressed in lymphatic endothelial cells.⁷⁸ It serves as a crucial lymphatic docking receptor for immune cells, such as DCs,⁷⁹ enabling their entry into peripheral lymphatics to prime adaptive immune cells such as CD8⁺ T cells.⁸⁰ LYVE1 can be enzymatically cleaved and released into the peripheral circulation,^{23,81} contributing to its presence in plasma. The observed elevated levels of LYVE1 may reflect cleavage and thus limited availability of LYVE1 on the lymphatic endothelium, significantly hindering DC migration and T cell priming, as demonstrated in an animal study using a *Lyve1*^{−/−} mouse model or LYVE1 neutralizing antibodies.⁸² Although these findings suggest a possible mechanism by which impaired DC trafficking could contribute to reduced adaptive immune responses in ME/CFS, our data cannot establish a causal link and thus warrant validation by future studies. Given the role that CD4⁺ and CD8⁺ CD27⁺CD28⁺ terminal effector T cells play in immune responses characterized by the production of granzyme B, perforin, and cytokines,^{77,83} the low proportion of these cell subtypes has been implicated in the ME/CFS patients' reduced capacity to resolve infection.⁸⁴ This limited response by adaptive immune cells correlates well with the reduced effector adaptive populations in our patient cohort. These intricate and complex relationships shed light on potential mechanisms contributing to the clinical symptoms observed in ME/CFS patients, particularly fatigue and susceptibility to infection.

We did not observe any differences in pro-inflammatory cytokines between cohorts in our study, and this is consistent with some reports but differs from others, reflecting the inconsistent detection of cytokines in ME/CFS plasma, as summarized in our previous comprehensive review.¹³ The lack of elevated pro-inflammatory cytokines could be attributable to the combination of reduced frequency of mature immune cells, as discussed above, or actions of other plasma proteins elevated in the patient cohort, namely PRG4 and PROC. PRG4 is a mucinous glycoprotein with a potent anti-inflammatory effect. By inhibiting the activation of toll-like receptor 2/4 signaling pathways and CD44, it plays a key role in maintaining an anti-inflammatory macrophage phenotype³⁵ as well as limiting neuroinflammation.³⁴ PROC is a zymogen secreted by the liver and becomes activated after binding to the PROC receptor.⁸⁵ Despite being predominantly known for its anticoagulant activity, PROC also has anti-inflammatory functions mediated by reducing the synthesis of nuclear factor κβ (NF-κβ) components and their nuclear translocation.³⁶ This in turn reduces the transcription of cytokine genes. In previous studies, animals with PROC deficiency develop prothrombotic and inflammatory phenotypes^{86,87} but can be rescued with the addition of recombinant human-activated PROC.⁸⁷

Our proteomic data revealed elevated levels of complement cascade proteins CFHR2, CFHR5, and MASP1 but not any of the major components of the pathway as quantified by previous studies.^{88–90} The presence of CFHR2 and CFHR5 appears to constitute a feedback mechanism with negative regulation of the major complement pathway proteins^{29,30} while MASP1 represents a preference for the immunoglobulin-independent MBL

pathway.^{38,39} While these observations are consistent with a potential feedback mechanism, further studies are needed to clarify their relevance in ME/CFS.

Several of the differentially expressed proteins in the plasma may be relevant to ME/CFS pathophysiology. Among these, soluble THBS1, VWF, and VASN are implicated in vascular reactivity, potentially contributing to the endothelial dysfunction observed in ME/CFS patients.^{91,92} Soluble THBS1 promotes platelet aggregation at the site of vascular injury in the presence of VWF,²⁸ which was also elevated in our study. This interaction facilitates thrombus formation.^{27,93} Furthermore, THBS1 can bind to two receptors, CD36 and CD47, to antagonize the nitric oxide signaling pathway. Binding to these receptors on vascular smooth muscle cells impairs vasodilation⁹⁴ and restricts blood flow, while when binding to endothelial cells, THBS1 inhibits cell growth and angiogenesis.²⁵ These activities could collectively reduce overall blood flow in the ME/CFS patient and may contribute to the severity of fatigue experienced by these individuals.⁹⁵ This finding of elevated levels of THBS1 corroborates an earlier study suggesting that this antiangiogenic glycoprotein restricts brain-blood flow in a subset of ME/CFS patients, possibly contributing to brain fog and post-exertional malaise.⁹⁶ This has prompted the initiation of the STOP-ME study funded by Open Medicine Foundation Canada, aiming to investigate THBS1's role in ME/CFS pathogenesis. Additionally, THBS1 has been shown to suppress DC activation and secretion of cytokines,²⁶ which could explain the significant decrease of cDC2 populations in our patient cohort and the minimal differences in the inflammatory cytokines between the cohorts observed in our study. VASN is a transmembrane glycoprotein that exerts its function by inhibiting the activation of the transforming growth factor β (TGF- β) pathway. This occurs when its secreted form binds to TGF- β , preventing the growth factor from interacting with its receptor.²⁴ The TGF- β signaling pathway plays a critical role in the formation of vascular plexus and integrity.⁹⁷ A previous study reported impaired vascular development with hyperdilated, leaky vessels in mice lacking TGF- β signaling pathway components.⁹⁸ In view of a similar impact on vascular integrity to THBS1, this protein may also contribute to brain fog in ME/CFS patients. While FN1 may not have a direct role in vascular modeling, it can promote aggregation of platelets and thrombus formation,³³ thus possibly contributing to the restriction of blood flow and warranting further study. Additionally, the significantly lower amount of soluble CDH5 in the plasma in the ME/CFS patient cohort supports the notion of altered vascular integrity in ME/CFS. CDH5 is a critical protein for vasculogenesis⁹⁹ and blood vessel maintenance,¹⁰⁰ as evidenced by studies demonstrating severe vascular malformations in CDH5^{-/-} mice¹⁰¹ and venous stasis in mice treated with anti-CDH5 antibody.¹⁰² The reduced presence of shed CDH5 in plasma may indicate increased retention of this protein at endothelial junctions, possibly reflecting compensatory mechanisms for vascular remodeling or repair in ME/CFS patients. However, although these plasma proteins are involved in vascular regulation, their precise roles in ME/CFS remain unproven and will require confirmation in future studies.

We observed lower overall levels of multiple immunoglobulin isotypes in ME/CFS patients' plasma. It remains to be deter-

mined whether these immunoglobulin differences contribute to ME/CFS pathophysiology. The elevated levels of particular immunoglobulin variable domains could reflect prior exposure or ongoing immune responses to organisms such as human parainfluenza virus 3, which is associated with IGHV5-51.⁴³ However, our study does not provide direct evidence for such association, and further research are needed to examine whether such specific immune responses occur in ME/CFS patients. Similar antibody production biases were previously reported¹⁰³; however, differences in the technical aspects of sample preparation and instrument sensitivity preclude a direct comparison with our data.

Taken together, the molecular and immune cell changes observed in our study complement the canonical pathways identified by IPA, namely downregulation of immune-related signaling and upregulation of vascular and integrin-related pathways. Although pathway enrichment analysis does not provide direct causal evidence, it highlights the biological relevance of these findings in ME/CFS. It is noteworthy that our pathway enrichment of RAF/MAPK signaling in plasma is consistent with PBMC transcriptomic enrichment studies,^{74,104,105} suggesting this pathway may be of importance in ME/CFS. In contrast, B-cell-related pathways were downregulated in our plasma enrichment analysis but upregulated in PBMC enrichment,⁷⁴ likely reflecting differences between soluble proteins and cellular transcriptional programs. Although no enrichment analysis was performed in a separate MS-based plasma proteomics study,¹⁰⁶ their findings of vascular and endothelial dysfunction complement our enrichment of vascular and integrin-related pathways. Future work is needed to validate these pathway-level associations across ME/CFS cohorts.

Finally, our exploratory analysis identified a model comprising seven variables that was associated with ME/CFS and suggested predictive potential to distinguish ME/CFS patients with 85.2% sensitivity, 96.7% specificity, and 91% accuracy. Notably, these variables span adenosine metabolism (AMP), immune functions (cDC1, LYVE1, and IGHG2), and vascular factors (FN1, VWF, and THBS1), highlighting potential interactions between these areas of dysfunction that contribute to ME/CFS. This exploratory model performs comparably to the established questionnaire DSQ-SF with 87.7% sensitivity, 84.7% specificity, and 86.8% accuracy in classifying individuals by the Canadian Consensus Criteria.¹⁴ While further validation in an independent cohort of ME/CFS patients is essential, these findings suggest that our model could support future diagnostic advances. Accurate diagnosis using such a model may reduce diagnostic delays, alleviating the prolonged suffering and economic burden faced by patients. ME/CFS imposes substantial economic costs, with annual expenses estimated at \$63,000 per patient, primarily due to indirect costs such as lost productivity and work absence.¹⁰⁷ Although there may be an initial cost in implementing this tool, its potential to reduce these indirect costs and improve patient quality of life makes it a worthwhile consideration.

Sex-stratified analysis revealed that female ME/CFS patients exhibited reduced ATP levels and ATP/ADP ratios in PBMCs, alongside elevated AMP in plasma and PBMCs. Reduced plasma adenosine in female patients was also observed in a

previous metabolomic study,¹⁰⁸ suggesting impaired ATP synthesis or turnover. Additionally, a lower proportion of cDC2, CD56^{low}CD16⁺ NK cells, CD27⁺CD28⁺ CD4⁺ terminal effector memory, and CD27⁺CD28⁺ CD8⁺ intermediate effector memory T cell subsets was observed in female patients compared to healthy controls. The plasma proteomic profile in female patients was broadly similar to that of the overall cohort, although not all protein changes were identical. Importantly, the five proteins identified as the most significant contributors to predicting ME/CFS were also observed in the female subgroup. Overall, these findings may be due to estrogen-mediated regulation of mitochondrial and immune function,¹⁰⁹ as reported in other female-biased conditions such as multiple sclerosis.¹¹⁰ However, due to the limited number of male participants, we were unable to perform a reliable comparison between male ME/CFS patients and healthy male controls or to directly contrast male and female ME/CFS subgroups.

In conclusion, this study provides compelling evidence that ME/CFS is associated with dysfunction across multiple biological systems, challenging its dismissal as a psychological disorder. Notably, we identified variables from three distinct biological systems with strong predictive potential for ME/CFS, highlighting the crosstalk between immune, vascular, and energy production dysfunction. These findings reinforce the legitimacy of ME/CFS as a medical condition and should pave the way for promoting broader acceptance of the condition within the medical community and society at large, thus improving patient management and outcomes.

Limitations of the study

Our study has certain limitations. First, our patient cohort did not stratify according to disease severity; therefore, the association between dysfunction and severity could not be determined. A notable feature of the cohort demographics is the predominance of women, typical of the sex distribution of ME/CFS patients¹¹¹ (Table 1). Consequently, the small number of male healthy controls and patients results in insufficient statistical power for robust analysis in this subgroup. Hence, statistical analysis was performed on the entire cohort and on female healthy controls and patients. Next, patients recruited to this study were not sampled at a uniform time point; samples could not be captured at the start of ME/CFS but rather after varying durations of the condition. Additionally, the heterogeneous clinical status of the patients includes nearly 20% who are disabled (i.e., score ≤ 50), yet 44.3% who are capable of a degree of normal activity (score ≤ 70). Hence, there is a potential for confounding by other underlying conditions that may affect ME/CFS. This shortcoming could potentially be addressed through longitudinal sampling to determine the stability or fluctuation of measured parameters over time. Without identifying the underlying cause of ME/CFS, a therapeutic target cannot be determined. Nonetheless, there may be scope for targeting known pathways to provide symptomatic relief. The analytical chemistry (LC-MS/MS, GC-MS, uHPLC, and HPLC) and flow cytometry (conventional and spectral) methods used here, while validated for plasma and PBMCs, have inherent limitations. Some metabolite and cytokine measurements were close to the lower limits of detection, reducing precision and potentially introducing variability that could affect

statistical analyses and the reliability of certain results. The plasma proteome approach may also underrepresent low-abundance proteins of potential relevance to ME/CFS.¹¹² Future studies could improve coverage by depleting high-abundance proteins or applying targeted approaches such as recombinant protein spectral libraries.^{113,114} Similarly, the 25-color flow cytometry panel available at the time limited resolution of minor subsets (exhaustion markers or MAIT cells). However, there are 40-color panels available now that permit deeper immune profiling. Although our study benefits from a relatively large and well-matched cohort, effect sizes for some variables remain modest, and validation in independent cohorts will be essential to confirm reproducibility and clinical relevance. This limitation has also been noted in another recent multimodal study of ME/CFS,⁷⁴ which, despite important insights, was constrained by a smaller cohort size. Table S3 contains the effect sizes and corresponding sample sizes (80% power) for the most significant and biologically relevant findings using Cliff's δ .

RESOURCE AVAILABILITY

Lead contact

Request for further information and resources should be directed to and will be fulfilled by the lead contact, Dr. Benjamin Heng (benjamin.heng@mq.edu.au).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- Proteomics data generated in this study are available through ProteomeXchange (PRIDE). The accession number is listed in the deposited data section of the [key resources table](#).
- Raw metabolomic, immune, and demography data are stored on Figshare with restricted access. These data cannot be deposited in an external public repository because participant consent for open sharing was not obtained at recruitment. Participant-level data can be requested from the lead contact with a detailed research proposal outlining the intended use of the data. Data access will only be provided once an appropriate Human Research Ethics Committee (HREC) or equivalent institutional approval is in place.
- This paper does not report original codes.
- Any additional information required to reanalyze the data reported in this work is available from the [lead contact](#) upon request.

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AUTHOR CONTRIBUTIONS

B.H., S.B., R.S., and G.J.G., conceived and supported the study; P.P.H., and A.A.B., performed statistical analysis; B.H., S.B., S.C., A.S.P., and G.J.G.,

metabolomic sample processing and data acquisition; B.H., S.B.A., D.P.T., and S.K., proteomic sample processing and data acquisition; B.H., B.G., S.K., A.S., and M.P.-H., spectral flow cytometry cells processing and data acquisition; B.H., B.G., and S.K., cytokine assay processing and data acquisition; R.S., Y.M., and K.S., recruitment, clinical information, and sample collection from participants; B.H., S.B.A., B.G., S.K., and S.C., clinical sample processing; B.H., S.B.A., B.G., S.B., M.P.-H., S.C., A.S.P., G.J.G., S.K., A.S., and R.S., data analysis and interpretation; B.H. wrote the manuscript, with all authors having read and approved the final version of the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.xcrm.2025.102514>.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Chemicals, peptides, and recombinant proteins		
Ficoll-Paque Plus	Merck	Cat# GE17-1440-02
PBS	Gibco	Cat# 10010023
RPMI-1640	Gibco	Cat# 11875093
DMSO	Thermo Scientific	Cat# J66650.AE
Trichloroacetic acid	Sigma Aldrich	Cat# T6399
1,1,1,3,3,3 hexafluoroisopropanol	Sigma Aldrich	Cat# 105228
Trifluoroacetic anhydride	Sigma Aldrich	Cat# 106232
Toluene	Sigma Aldrich	Cat# 244511
Sodium bicarbonate	Sigma Aldrich	Cat# S6014
L-kynurenine	Sigma Aldrich	Cat# K8625
Kynurenic acid	Sigma Aldrich	Cat# K3375
L-tryptophan	Sigma Aldrich	Cat# 93659
2,3-Pyridinedicarboxylic acid	Sigma Aldrich	Cat# P63204
Picolinic acid	Sigma Aldrich	Cat# P5503
Anthranilic acid	Sigma Aldrich	Cat# A89855
3-Hydroxy-DL-kynurenine	Sigma Aldrich	Cat# H1771
3-Hydroxyanthranilic acid	Sigma Aldrich	Cat# H9391
Zinc acetate dihydrate	Sigma Aldrich	Cat# 96459
Sodium Acetate	Sigma Aldrich	Cat# 241245
Acetonitrile	Sigma Aldrich	Cat# 34998
2-Picolinic-d ₄ acid	Sigma Aldrich	Cat# 615757
Quinolinic acid-d ₃	MedchemExpress	Cat# HY-100807S
Formic acid	Fisher Chemical	Cat# A117
Ammonium acetate	Sigma Aldrich	Cat# 238074
Trypan Blue solution	Sigma Aldrich	Cat# T8154
DL-Dithiothreitol	Sigma Aldrich	Cat# D0632
Iodoacetamide	Sigma Aldrich	Cat# I1149
Trypsin	Promega	Cat# V5111
Adenosine 5'-monophosphate	Sigma Aldrich	Cat# 01930
Adenosine 5'-diphosphate	Sigma Aldrich	Cat# 01905
Adenosine 5'-triphosphate	Sigma Aldrich	Cat# 7699
β-Nicotinamide adenine dinucleotide, reduced	Sigma Aldrich	Cat# N8129
β-Nicotinamide adenine dinucleotide hydrate	Sigma Aldrich	Cat# 43410
β-Nicotinamide adenine dinucleotide 2'-phosphate reduced	Sigma Aldrich	Cat# N6505
β-Nicotinamide adenine dinucleotide phosphate hydrate	Sigma Aldrich	Cat# N3886
Nicotinamide-d ₄	Toronto Research Chemicals	Cat# N407752
β-Nicotinamide adenine dinucleotide, reduced-d ₅ (d ₄ -major)	Toronto Research Chemicals	Cat# N201487
β-Nicotinamide Adenine Dinucleotide-d ₄ Ammonium Salt (d ₃ major)	Toronto Research Chemicals	Cat# N407783
β-Nicotinamide-d ₄ Mononucleotide (d ₃ -major)	Toronto Research Chemicals	Cat# N407768
Adenosine 5'-Diphosphate-13C ₅	Toronto Research Chemicals	Cat# A281697
Adenosine 5'-Monophosphate-13C ₅	Toronto Research Chemicals	Cat# A281782
Adenosine-13C ₁₀ ,15N ₅ 5'-triphosphate disodium salt solution	Sigma Aldrich	Cat# 645702
FBS	Bovogen	SFBS-AU

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Continued		
REAGENT or RESOURCE	SOURCE	IDENTIFIER
Critical commercial assays		
LEGENDPlex Inflammation Panel 1 (13-plex)	BioLegend	Cat# 740007
Human B Effector 2 Panel (5-plex)	BioLegend	Cat# 740531
Immunoprofiling Assay, cFluor® Reagent Kit (18 colors)	Cytek	Cat# SKU R7-40002
Immunoprofiling Kit, 7 Color (Brilliant Violet™)	BioLegend	Cat# 900004160
Pierce™ BCA Protein Assay Kits	Thermo Scientific	Cat# 23225
Deposited data		
Proteomic data	Proteomexchange	PXD068504
Metabolites, cytokine, flow cytometry and demography data	Figshare	https://doi.org/10.25949/30195544
Software and algorithms		
Agilent OpenLAB CDS ChemStation (Edition C.01.04)	Agilent	https://www.agilent.com/en/product/software-informatics/analytical-software-suite/chromatography-data-systems/openlab-chemstation
Agilent GC/MSD ChemStation software (Edition 02.02.1431)	Agilent	https://www.agilent.com/en/product/software-informatics/analytical-software-suite/chromatography-data-systems/openlab-chemstation
SpectroFlo Software (V2.2.0.2)	Cytek	https://cytekbio.com/pages/spectro-flo
Spectronaut	Biognosys	https://biognosys.com/software/spectronaut/
Ingenuity Pathway Analysis	Qiagen	https://digitalinsights.qiagen.com/products-overview/discovery-insights-portfolio/analysis-and-visualization/qiagen-ipa/
Prism 10 (Version 10)	GraphPad	https://www.graphpad.com
IBM SPSS Statistics (Version 29)	IBM	https://www.ibm.com/products/spss-statistics
JMP Software (Version 17.2)	IBM	https://www.jmp.com/en/home
FlowJo Software (Version 10.8.2)	BD Bioscience	https://www.bdbiosciences.com/en-ch/products/software/flowjo-software?tab=flowjo-v10-software
R studio	Posit	https://posit.co/download/rstudio-desktop/
LEGENDPlex Data Analysis Software	BioLegend	https://www.biolegend.com/en-ie/immunoassays/legendplex/support/software

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Study cohorts

Patients were recruited from The Grove Health Pymble clinic in Sydney, Australia. Potential patients were clinically diagnosed with ME/CFS as defined by the Canadian Consensus Criteria¹ and satisfied the case definition classification according to the DSQ-SF.¹⁴ Additional inclusion criteria were age ≥ 18 , adequate understanding of English and the ability to provide informed consent, a low score in physical status (physical functioning score from SF-36,¹¹⁵ and Karnofsky Performance Scale¹⁵ of ≤ 70). The exclusion criteria consisted of pregnancy, inability to provide informed consent, and not carrying a clinical diagnosis of ME/CFS as defined by the Canadian Consensus Criteria.¹ The healthy control group was selected to approximately match the age and sex distribution of the ME/CFS patient cohort. Informed consent was obtained from all participants, and our study was approved by the Macquarie University Human Research Ethics committee (ref. 52024922255391). Recruitment was randomised and concurrent across both cohorts to avoid collection bias.

METHOD DETAILS

Data collection from questionnaires

Health status and quality of life were assessed on the day of physical examination and biological sample collection from patients and healthy control subjects using the DSQ-SF, SF-36 and Karnofsky Performance Scale. The DSQ-SF comprises 14 questions assessing the frequency and severity of major ME/CFS symptoms. It employs a “2-2” threshold for both frequency and severity whereby the presence of a symptom is determined when both values are “2”. The SF-36 evaluates various aspects of physical and social functioning, physical and emotional limitations, energy, pain, general health and health changes. The Karnofsky Performance Scale assesses the functional and performance status of the participant, scored between 10 and 100 and was administered by a clinician.

Human peripheral blood collection

Blood samples were collected from non-fasting participants into BD Vacutainer EDTA tubes (BD Biosciences, Cat#366473) by a registered nurse or clinician, and processed within an hour to minimise any storage effects.¹¹⁶ Fasting was not required as it could exacerbate symptoms in some ME/CFS participants. Plasma was separated and collected from whole blood after initial centrifugation at 1300×g for 10 min at 20°C with no brake. Next, the remaining components of the blood were diluted with an equal volume of phosphate-buffered saline (PBS; Thermo Fisher Scientific, MA, USA) before being layered on top of Ficoll-Paque Plus (Sigma Aldrich, MO, USA) and centrifuged at 400×g for 40 min at 20°C with no brake. The PBMC layer was transferred into a new tube, washed with PBS, and the cell pellet was resuspended with freezing medium [consisting of RPMI-1640 (Thermo Fisher Scientific, MA, USA) supplemented with 10% human serum and 10% DMSO (Thermo Fisher Scientific, MA, USA)] for storage. Plasma and PBMC were cryopreserved in liquid nitrogen and stored for a maximum of two months until sufficient numbers of healthy controls and ME/CFS patients were enrolled. To minimise potential effects of repeated freeze–thaw cycles, samples were aliquoted for individual assays, and each aliquot underwent only a single thaw cycle prior to analysis. Cell quantity and viability were assessed using Trypan Blue staining. One of the healthy control blood samples did not yield sufficient cell numbers for NADomics, adenine nucleotide quantification, and/or immune profiling.

Sample preparation for targeted metabolomic analysis

1 volume of 10% (w/v) trichloroacetic acid (Sigma Aldrich, MA, USA) was added to deproteinise plasma samples before centrifugation at 12,000×g for 10 min at 4°C. The plasma supernatants were collected and filtered through a 0.22 µm PTFE syringe filter (Merck-Millipore, CA, USA) before transferring into vials for liquid chromatography analysis.

For gas chromatography analysis, an aliquot of deproteinised plasma was mixed with a fixed concentration of deuterated internal standards for QUIN and PIC. The mixture was dried under vacuum and derivatized with 120 µL of 1,1,1,3,3,3 hexafluoroisopropanol and trifluoroacetic anhydride for 1 h at 60°C. Toluene was used to extract the fluorinated esters and washed with 5% sodium bicarbonate. Finally, the esters were dried using sodium sulfate packed pipette tips and then transferred into vials for gas chromatography analysis.

KP metabolite quantification by high-performance liquid chromatography (HPLC)

The Agilent 1290 Infinity HPLC (Agilent, CA, USA) was used to determine the plasma levels of KP metabolites, TRP, KYN, 3HK, 3-HAA and AA as previously described.¹¹⁷ 20 µL of processed samples were injected into an analytical column ZORBAX Rapid Resolution High Definition C18 (2.1 mm × 150 mm, 1.8µm particle size; Agilent, CA, USA) by reverse phase gradient elution with an isocratic flow rate of 0.75 mL/min for 12 min at 38°C. The mobile phase used was 0.1 M sodium acetate (pH 4.6). TRP was detected at excitation/emission wavelength of 280 nm/438 nm while 3HAA and AA were detected at excitation/emission wavelength of 320 nm/438 nm using the fluorescence detector (G1321B; xenon flash lamp; Agilent). The UV detector set (G4212A; Agilent) was used to detect KYN and 3HK at 365 nm (reference signal off). The chromatogram output of these KP metabolites was analyzed using the Agilent OpenLAB CDS ChemStation (Edition C.01.04).

The Agilent 1260 Infinity HPLC (Agilent, CA, USA) was used to determine the KYNA plasma levels as previously described.¹¹⁷ 10 µL of processed samples were injected into an analytical column ZORBAX Rapid Resolution High Definition C18 (4.6 mm × 100 mm, 3.5 µm particle size, Agilent Technologies, CA, USA) using a reverse phase gradient elution with an isocratic flow rate of 1 mL/min for 8 min at 30°C. The mobile phase used was 95% of 0.05 M sodium acetate, 0.05M zinc acetate (pH 5.2), and 5% HPLC grade acetonitrile. The fluorescence detector (G1321B xenon flash lamp, Agilent, CA, USA) was set at excitation/emission wavelength of 344 nm/388 nm for detection of KYNA. The chromatogram output of KYNA was analyzed using the Agilent OpenLAB CDS ChemStation (Edition C.01.04).

KP metabolite quantification by gas chromatography-mass spectrometry (GCMS)

The plasma levels of PIC and QUIN were quantified using the Agilent 7890 GCMS (Agilent, CA, USA) coupled with an Agilent 5975 mass spectrometer as previously described.¹¹⁷ 1 µL of processed sample was injected using spitless mode onto an HP-5MS GC capillary column (Agilent, CA, USA). Analysis was performed in negative chemical ionization mode. Selected ions (m/z 273 for PIC, m/z 277 for d4-PIC, m/z 467 for QUIN and m/z 470 for d3-QUIN) were simultaneously monitored. GC oven was set at 75°C for 3 min, before ramping up to 300°C at a rate of 25°C/min and maintained at 300°C for 4 min for a total run time of 15.6 min.

Calibration curves of QUIN and PIC were constructed using peak area ratios (peak area of the QUIN and PIC divided by peak area of d3-QUIN and d4-PIC, respectively) of each calibrating solution versus its concentration. All spectra were processed, and peak areas integrated using Agilent GC/MSD ChemStation software (Edition 02.02.1431).

NADomics and adenine nucleotide quantification by liquid chromatography/MS-MS (LC-MS/MS)

The levels of AMP, ADP and ATP (in plasma and PBMC) and NAD, NADH, NADP, NADPH (in PBMC) were quantified using the TSQ Vantage MS (Thermo Fisher, MA, USA) coupled with the Vanquish (Thermo Fisher, MA, USA) solvent delivery/autosampler system. 20 μ L of processed sample were injected into an amino (NH₂) column (150 mm $\times$ 2 mm, 3 μ m, 100 Å, Phenomenex, Torrance, CA, USA) using a gradient elution at 25°C with 5 mM aqueous ammonium acetate and acetonitrile gradient as previously described.¹¹⁸ For normalization, all PBMC samples were adjusted to 2 million cells prior to extraction. One ME/CFS patient sample failed to generate signals for quantification. In addition to a healthy control blood collection that had insufficient immune cells mentioned under sample collection, this resulted in analysis of 60 healthy controls and 60 patient samples. For the calculation of NADP/NADPH ratio, only 47 of the 60 controls were included as 13 controls recorded zero concentration for NADP.

Analysis of plasma cytokines

LEGENDPlex Inflammation Panel 1 (13-plex, Cat: 740007, BioLegend) and Human B Effector 2 Panel (5-plex, Cat: 740531, BioLegend) assays were performed to quantify the levels of 13 and 5 human cytokines/chemokines respectively in patient and healthy control plasma. The assays were performed according to the manufacturer's manual, using a BD LSR Fortessa cell analyser (BD Bioscience, NJ, USA). Data analysis was performed using the LEGENDPlex Data Analysis Software (BioLegend, CA, USA). IL-6 and TNF measurements were included in both kits. Therefore, the averaged values of these cytokines were used for statistical analysis.

Spectral flow cytometry preparation and data analysis

PBMC were stained for spectral cytometry in three batches using the 25-Color Immunoprofiling Assay, cFluor Reagent Kit with 18 colors (Cat: SKU R7-40002, Cytex) and Immunoprofiling Kit, 7 Color (Brilliant Violet) (Cat: 900004160, BioLegend) according to the manufacturer's protocol. Cell viability was assessed after thawing using Trypan Blue staining to determine the proportion of live cells prior to staining (Table S4). Each batch contained a representation of healthy control and patient groups and a batch control PBMC sample from a healthy donor. All antibodies were validated, titred, and supplied in per-test amounts by Cytex. Single stained reference controls and gating strategy (Figure S6) were performed according to the manufacturer's instructions.

Stained cells were analyzed using a Cytex Aurora flow cytometer (5-laser; 355 nm, 405 nm, 488 nm, 561 nm, and 640 nm) with SpectroFlo Software v2.2.0.2. The raw fluorescent signals were unmixed using appropriate reference controls, with autofluorescence extraction enabled. To reduce unmixing errors, the spectrum of the single stained reference controls was verified against the expected spectrum profile provided by Cytex Full Spectrum Viewer (<https://spectrum.cytexbio.com>). The unmixed FCS files were used for further data analysis using FlowJo 10.8.2 (Becton Dickinson & Company). Single events were distinguished from doublets and debris, and ViaDye-negative live cells were then selected. Identification of major PBMC populations was performed using Cytex's gating strategy which accompanied the reagent kit (Figure S6). Analysis of cell subset percentages included both percent of live cells (% live) and/or percent of parent subset (% parent). Percentages of live cells (% live) were derived from the live cell gate. Further analysis was also performed on R using Spectre,¹¹⁹ to correct for batch alignments,¹²⁰ clustering,¹²¹ generating dimensionality reduction (t-SNE) plots¹²² (Figure S6).

Sample preparation for proteomics

Plasma protein concentrations were measured using a BCA Protein Assay Kit (ThermoFisher, MA, USA) following the manufacturer's protocol. The protein samples underwent a reduction step using 15 mM dithiothreitol at 60°C for 30 min, after which they were alkylated with 30 mM iodoacetamide at room temperature for 30 min while shielded from light. Following that, the samples were digested with trypsin at a 1:30 ratio for 16 h at 37°C with gentle agitation. In preparation for LC-MS analysis, the digested peptide mixtures were subjected to desalting and purification using the C18 StageTips method.¹²³

Protein analysis by LC-MS

The desalted samples were resuspended into loading buffer (0.1% FA) and analyzed by liquid chromatography-mass spectrometry (LC-MS). Peptides were separated using a Nexera Series UHPLC LC-40 system (Shimadzu, Kyoto, Japan) coupled to an InfinityLab Poroshell 120 Å, 1.9 μ m, 2.1 mm $\times$ 50 mm C18 column with a UHPLC guard attached (Agilent, Santa Clara, United States) maintained at 40°C. LC mobile phase buffers were comprised of A: 0.1% (v/v) formic acid, 99.9% water and B: 0.1% (v/v) formic acid, 99.9% acetonitrile. The peptides were eluted using a linear gradient of 9% B to 29% B over 5.6 min, then 29% B to 42% B over 0.6 min, then 42% B to 80% B over 0.2 min. The column was washed with 80% B for 0.2 min before re-equilibrating with 3% B. The flow rate was 0.8 mL/min.

The separated peptides were analyzed with a ZenoToF 7600 (Sciex, Toronto, Canada) with the sequential window acquisition of all theoretical (SWATH) MS experiment conducted in positive polarity. Ion source gases were: ion source 1:60 psi, ion curtain gas at 40 psi, and CAD gas at 7 psi. Spray voltage was set to 4500 V and the heated capillary set to 700°C. Ions were generated by electrospray

and data acquired in SWATH acquisition mode using varying isolation widths of approximately 5–42 m/z (containing 1 m/z for the window overlap and decreasing in width toward 650 m/z). A set of 60 overlapping windows was constructed covering the precursor mass range of 400–900 m/z and the effective isolation windows can be considered as being 399.5–441.2, 440.2–459.9, etc. SWATH MS2 spectra were collected from 100 to 1800 m/z. The collision energy for each window varied 19–43 V, starting at 19 V and increasing 1–2 V per window with increasing m/z. An accumulation time of 5 ms was used for all fragment-ion scans in high-sensitivity mode, resulting in a total duty cycle of ~ 0.627 s. Zeno pulsing was applied.

Raw DIA-MS files were analyzed using Spectronaut (Biognosys) with the DirectDIA, library-free workflow. Peptide lengths ranged from 9 to 52 amino acids, with enzyme specificity set to trypsin, allowing a maximum of one missed cleavage site. Carbamidomethylation of cysteine (C) was applied as a fixed modification, while oxidation of methionine (M) and N-terminal acetylation were included as variable modifications, allowing up to five modifications per peptide. At least one unique peptide per protein was selected for quantification. The false discovery rate (FDR) was set at 0.01 (1%) for both peptides and proteins.

Ingenuity pathway analysis

Ingenuity Pathway Analysis (IPA; Qiagen, Redwood City, CA, USA) was used to identify enriched biological pathways and functions associated with differentially expressed proteins (FDR < 0.05) through canonical pathway and functional annotation analyses. Statistical significance of pathway enrichment was determined using Fisher's exact test ($p < 0.05$), while activation or inhibition states were inferred using IPA's activation Z score, with thresholds of $\geq +2$ or ≤ -2 considered significant.

QUANTIFICATION AND STATISTICAL ANALYSIS

Statistical analysis

t test and Pearson's Chi-squared test were used to analyze the difference of age and sex between the cohorts respectively. The distribution of metabolite data was assessed using Shapiro-Wilk test, while the proportion of cell subsets from flow cytometry were logit-transformed before their distributions were evaluated using the Shapiro-Wilk test. The Mann-Whitney test was applied to non-normally distributed data whereas Welch's *t* test was used for normally distributed data. Effect sizes were calculated using Cliff's δ as nonparametric Wilcoxon rank-sum test was used to detect group difference. Correlations between plasma and PBMC adenosine metabolites were assessed using Spearman's correlation. The differences between the compared groups of values were considered to be statistically significant at a p -value ≤ 0.05 . Graphs were generated using GraphPad Prism 10 (GraphPad, San Diego, USA).

Quantitative MS proteomics analysis results were first filtered for protein groups that had observations in at least 50% of the cohorts. This threshold was chosen to retain proteins that may be specifically expressed in either the disease or control group but missing in the other. Thus, this would result in these proteins being detected in approximately 50% of the total cohort. Exclusion of such proteins based on presence thresholds could lead to the loss of biologically relevance and/or condition-specific signals and potentially result in misleading conclusions. This approach aligns with common practices in omics studies, where a 50% "present call" threshold is often used in gene expression analyses to ensure the inclusion of informative features across cohorts.^{124,125} We have also included a 70% cohort observation threshold for increased stringency as a comparison. The data was then normalised by adjusting the median peak intensity of each sample to the global median. Missing values are a common occurrence in large-scale proteomics datasets and often arise from the inability to detect low-abundance proteins due to the sensitivity limitations of mass spectrometry. Missing values were imputed using a Missing Not At Random (MNAR) strategy, which assumes that missing values correspond to low-abundance proteins.¹²⁶ For each variable, missing values were randomly imputed around a value shifted below the mean of the observed intensities, with variability introduced based on the standard deviation of the observed data. This approach ensures that imputed values fall near the lower end of the observed intensity range, effectively mimicking low-abundance proteins.¹²⁷

Relative abundance comparison was performed across sample groups and analyzed by *t* test in R. Proteins with a p -value < 0.05 and a fold change > 1.2 were determined to be differentially expressed between healthy control and patient with ME/CFS. This cutoffs criterion, which combines fold change and p -value, has been validated in spike-in studies using DIA analysis that demonstrates to control the false discovery rate effectively.^{128,129}

Statistical modeling

CART modeling was used to develop a classification model using IBM SPSS Statistics version 29 and JMP version 17.2. All 114 biological parameters, including sex and age, were included in the modeling process (Table S2). To reduce the chance of overfitting to the data, a 10-fold cross-validation process was applied with binary splits for any given node. Furthermore, upon identification of the resultant best subset of predictors, CART modeling was additionally repeated after excluding the principally identified predictor (first split) in the identified best subset. This was done to explore whether any other predictors, or groups of predictors, may become important or collectively better predict ME/CFS diagnosis. The "goodness of fit" of each of the resultant CART models was assessed primarily on sensitivity, specificity and AUC. Results are provided for the model having the highest sensitivity, specificity and AUC, along with more parsimonious models with similar diagnostic values.